Arun Sekhar

Bachelor of Engineering in Electronics & Communication Engineering

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Professional Summary

Motivated Electronics and Communication Engineering student with hands-on experience in computer hardware servicing, embedded systems, software-hardware integration, and Arduino-based development. Seeking full-time opportunities in embedded systems, semiconductor, CyberSecurity, hardware validation, or software engineering.

Education

Year	Degree/Certificate	Institute	CGPA/Percentage
2023	HSC - Computer Science	Government Higher Secondary School, Kalkulam	76.66%
2023-2027	B.E. Electronics and Communication Engineering	Amrita College of Engineering and Technology, Erachakulam	7.95/10.00

Experience

- **Self-Employed** *2023 - Present*
  Part-Time Computer Technician Remote/Local
  - Repaired, assembled and upgraded 15+ desktop computers and laptops.
  - Diagnosed hardware and software issues to improve reliability and performance.
  - Installed Windows, device drivers, BIOS updates and productivity software.
  - Built custom PCs for gaming, office and multimedia workloads.
  - Provided preventive maintenance and technical support.

Projects

- **Smart Door Lock System with App Integration**
  Arduino, Bluetooth, Biometrics & Actuators
  - Designed and developed a smart door locking mechanism utilizing an Arduino microcontroller and a servo motor for physical actuation.
  - Integrated an HC-05 Bluetooth module to enable seamless wireless lock and unlock functionality via a dedicated mobile application.
  - Incorporated multi-factor authentication by implementing both fingerprint (biometric) and PIN verification methods for enhanced access security.
- **Retro FM – Android FM Radio Application (In Progress)** *Ongoing*
  Reverse Engineering & Native Interface
  - Developing an Android FM radio application for MediaTek-based devices using reverse engineering and native C++ (JNI).
  - Analyzed proprietary FM radio libraries (`.so`) using Ghidra to understand hardware communication and driver interactions.
  - Integrated Android Studio, Kotlin, CMake and JNI to interface with native FM libraries and device nodes.
  - Investigated Linux device drivers, Android HAL, and low-level hardware access for FM tuner functionality.
  - Debugged native code, build configuration, and Android system-level integration using Logcat and GDB.
- **Arduino-Based Home Automation System**
  Embedded Software-Hardware Integration
  - Designed a home automation prototype using Arduino, relay modules and sensors.
  - Integrated embedded software with hardware for appliance control.
- **Custom PC Assembly & Optimization**
  Hardware Diagnostics & System Configuration
  - Built and optimized custom desktop computers.
  - Configured BIOS, installed operating systems, drivers and application software.
  - Refurbished used PCs to improve performance and reliability.

Technical Skills

- **Programming Languages**: C (Basic), C++ (Basic), Python (Basic)
- **Embedded**: Arduino, Embedded C, Microcontrollers
- **Hardware**: PC Assembly, Hardware Diagnostics, BIOS Configuration, Circuit Debugging
- **Tools**: Arduino IDE, MATLAB, Git (Basic), MS Office
- **Operating Systems**: Windows, Linux (Basic)
- **Networking**: Basic Configuration and Troubleshooting

Relevant Coursework

- **Core Subjects**: Embedded Systems, Digital Electronics, C Programming, Python Programming, Cybersecurity (In progress)

Interests

Embedded Systems, Semiconductor Engineering, Automation, Software Development, Cybersecurity, CCNA